

MA 251: Abstract Algebra

Course Code: MA 251

Course Name: Abstract Algebra

L-T-P-C: 3-0-0-3

Students intended for: B.Tech.

Elective or Core: Discipline Elective for B.Tech. Mathematics and Computing

Prerequisites:

Mutual Exclusion: MA 548

Approval: 59th BoA

Course Contents

- **Unit I:** Binary operation and its properties, Groups: examples and basic properties, Subgroups, Finite Groups and Subgroups. [6 Lectures]
- **Unit II:** Cyclic groups, Order of a group, Permutation groups, Isomorphism theorems, Cayley's theorems. [6 Lectures]
- **Unit III:** Coset of a subgroup, Lagrange's theorem and its consequences, Normal subgroups, Quotient group, Internal direct products. [6 Lectures]
- **Unit IV:** Homomorphisms, Kernel Image of a homomorphism, Direct product of groups, Structure of finite abelian groups. [6 Lectures]
- **Unit V:** Rings: definition, examples and basic properties, Subrings, Zero divisors, Integral domains, Fields, Characteristic of a ring, Ideals, Prime ideal, Maximal ideals, Quotient rings. (6 Lectures)
- **Unit VI:** Ring Homomorphisms, Isomorphism theorems, Field of Quotients, Division Algorithm in Polynomial Rings and its consequences. [6 Lectures]
- **Unit VII:** Reducibility and Irreducibility tests, Unique factorization in $\mathbb{Z}[x]$, Irreducibles, Primes, UFD, PID and Euclidean domains. [6 Lectures]

Text Books:

1. Joseph Gallian, **Contemporary Abstract Algebra**, 7th Edition.
2. N. Herstein, **Topics in Algebra**, Wiley Eastern Ltd., 1975.

References:

1. M. Artin, **Algebra**, Prentice Hall of India, 1994.
2. J. B. Fraleigh, **A first Course in Abstract Algebra**, Narosa Publishing House, 2003.
3. Klima, Sigmon and Stitzinger, **Applications of Abstract Algebra with Maple and MATLAB**, 2nd Edition, 2006.